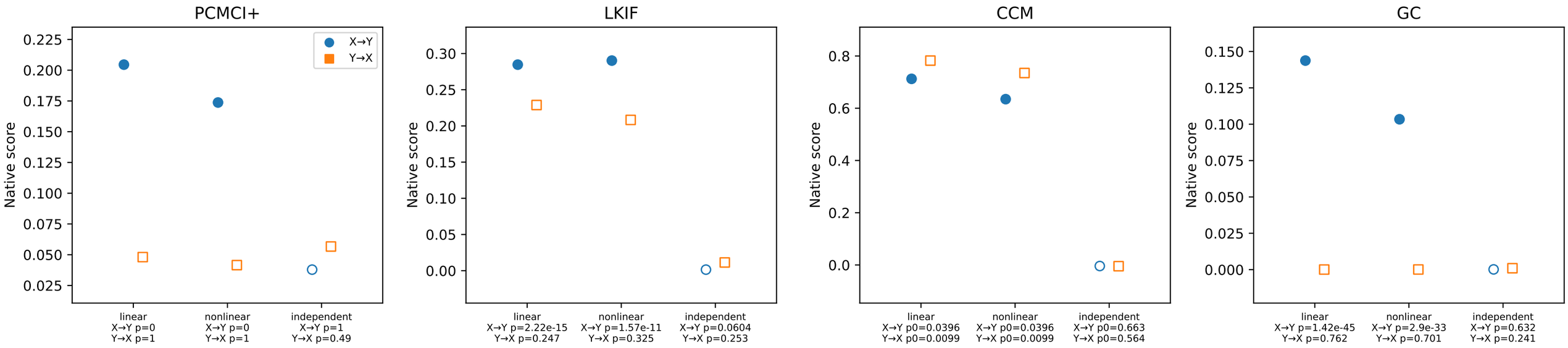
Baseline 2: irregular samples with accurate ages, then interpolation (20-yr grid)
Replicate 0; scores differ by method
PCMCI+
LKIF
CCM
GC
Native score
X→Y
Y→X
linear
X→Y p=0
Y→X p=1
nonlinear
X→Y p=0
Y→X p=1
independent
X→Y p=1
Y→X p=0.49
linear
X→Y p=2.22e-15
Y→X p=0.247
nonlinear
X→Y p=1.57e-11
Y→X p=0.325
independent
X→Y p=0.0604
Y→X p=0.253
linear
X→Y p0=0.0396
Y→X p0=0.0099
nonlinear
X→Y p0=0.0396
Y→X p0=0.0099
independent
X→Y p0=0.663
Y→X p0=0.564
linear
X→Y p=1.42e-45
Y→X p=0.762
nonlinear
X→Y p=2.9e-33
Y→X p=0.701
independent
X→Y p=0.632
Y→X p=0.241
Retained direction
Direction not retained
Unresolved evidence
CCM lag failed
Hollow markers include CCM lag-screen rejection. CCM score and p0 remain the original Tp=0 results.